

- Do not edit/delete any heading or section with **red text**.
- Type /paste you answers in **relevant sections (below Final Answer (in all questions , Approach (wherever required) and Charts & tables (wherever required))** of the solution
- Link to [submission guidelines](#)
- Please note that **submission not following the guidelines** will **not be accepted**

OBJECTIVE QUESTIONS

Question 1: What is the total no. of attributes present in the data?

Solution:

Final Answer (if any):

15

Approach (if any):

Counting all the unique attribute names - ID Ticket, Fecha, Employee ID, Agent ID, Request Category, Issue Type, Severity, Priority, Resolution Time (Days), Satisfaction Rate, Full Name, Email, Year of Birth, Month of Birth, Day of Birth

Charts & Tables (if any):

- Table / Chart Name:
<Name of the table or chart used>
- Description:
<What the table or chart represents>

Question 2: Which columns have inconsistent or missing values, and what is the count of such values?

Solution:

Final Answer (if any):

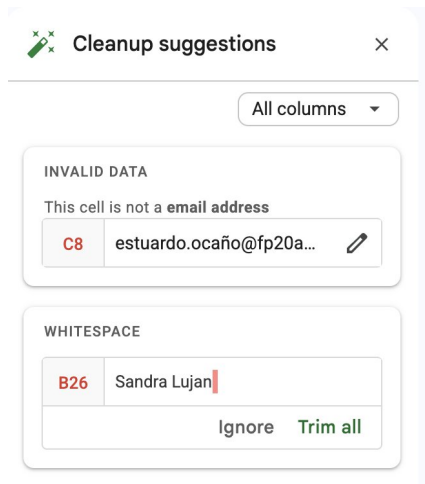
2 columns

Approach (if any):

Select data -> data -> data cleanup suggestions

Charts & Tables (if any):

Whitespace - IT Agents : Name column
Invalid data - IT Agents : Email column



Question 3: What is the average daily ticket volume over time?

Solution:

Final Answer (if any):

53.37

Approach (if any):

Pivot Chart -> Date(Fecha) in rows and COUNTA if Ticket ID in values gives us the tickets per date -> Calculate the Average of tickets per date.

Charts & Tables (if any):

Date	Number of tickets
01/01/2016	28
02/01/2016	43
03/01/2016	37
04/01/2016	36
05/01/2016	44
06/01/2016	33
07/01/2016	44
08/01/2016	39
09/01/2016	39
10/01/2016	30
11/01/2016	39
12/01/2016	40
13/01/2016	35
14/01/2016	29
15/01/2016	42
16/01/2016	38
17/01/2016	36
18/01/2016	34
19/01/2016	35
20/01/2016	43
21/01/2016	34
22/01/2016	31

Question 4: What is the distribution of ticket categories (e.g., Login Access, System, Software)?

Solution:

Final Answer (if any):

Hardware- 9.98% , Login Access- 29.94%, Software- 20.07%, System- 40.00%

Approach (if any):

Pivot Table -> Rows: Request Category , Values: Count of request category shown as percentage of grand total

Charts & Tables (if any):

Request Category	COUNTA of Request Category
Hardware	9.98%
Login Access	29.94%
Software	20.07%
System	40.00%
Grand Total	100.00%

Question 5: How many tickets has each agent handled?

Solution:

Final Answer (if any):

The Individual numbers are shown below using pivot chart

Agent ID	Agent Name	No. of tickets (per agent)
1	Mata Lucero	1969
2	JesusGrajeda	1968
3	Elena Velez	2021
4	Barraza Alberto	1988
5	Willyberto Gonzales	2000
6	A. Trejo	1949

7	Estuardo Ocaño	1935
8	Marisol Piedrahita	1960
9	Velasquez Jose	1949
10	Alberto Casillas	1974

Approach (if any):

Pivot Table(Tickets!A1:M97499)

Rows: Agent ID , Agent Name

Values: COUNTA of Ticket ID

Pivot table gives the No. of tickets handled by each agent identified by Agent Name and Agent ID

Charts & Tables (if any):

Agent ID	Agent Name	No. of tickets (per agent)
1	Mata Lucero	1969
2	JesusGrajeda	1968
3	Elena Velez	2021
4	Barraza Alberto	1988
5	Willyberto Gonzales	2000
6	A. Trejo	1949
7	Estuardo Ocaño	1935
8	Marisol Piedrahita	1960
9	Velasquez Jose	1949
10	Alberto Casillas	1974
11	Lopez Moran.	1956
12	Javier D.	1897
13	Griselda Galindo	1856
14	EstuardoTorres	1942
15	Galindo Guadalupe	1991
16	Orci Carlos	1926
17	Leon Lourdes	1961
18	Miller Gaviria	1892
19	Alfonso Barraza	1984
20	Eduardo Luna	1920
21	Alberto Gastelum	1889
22	Lorena	1966
23	Guadalupe Hernandez	1915
24	Barbara Grijalva	2003
25	Sandra Lujan	1906
26	Flores Sierra	1963
27	Isela Leyva	1968
28	Nurio Zepeda	1946

Question 6: How can you extract the domain from the email addresses in the IT Agents sheet?

Solution:

Final Answer (if any):

Using The formula : **MID('IT Agents'!C2,FIND("@",'IT Agents'!C2),(FIND(".", 'IT Agents'!C2, FIND(".", 'IT Agents'!C2)+1)-FIND("@",'IT Agents'!C2)))**

Approach (if any):

The formula uses **MID** to extract a piece of text from the email in 'IT Agents'!C2
FIND("@",'IT Agents'!C2) finds the position number where @ appears in the email
FIND(".", 'IT Agents'!C2) first finds the position of the first dot . in the whole email
FIND(".", 'IT Agents'!C2, FIND(".", 'IT Agents'!C2)+1) then finds the second dot . (the one before com) by starting the search after the first dot.

The difference **second_dot_position - @_position** gives how many characters lie between @ and that second dot (including the @)

MID(..., @_position, characters_between) finally returns the substring starting at @ and ending just before the dot before com

Eg: for lucero.mata@fp20analytics.com, the result is @fp20analytics.

Charts & Tables (if any):

- Table / Chart Name:
<Name of the table or chart used>
- Description:
<What the table or chart represents>

Question 7: How can you find the full name of an agent given their Agent ID?

Solution:

Final Answer (if any):

Using Lookup Formula

Approach (if any):

=VLOOKUP(\$D2,'IT Agents'!\$A\$1:\$H\$51,2,FALSE())

Charts & Tables (if any):

- Table / Chart Name:
<Name of the table or chart used>
- Description:
<What the table or chart represents>

Question 8: What is the count of each issue type (e.g., IT Error, IT Request)?

Solution:

Final Answer (if any):

IT Error - 24278

IT Request - 73220

Approach (if any):

Formula Used : =COUNTIF(Tickets!\$F\$2:\$F\$97499,"IT Error")

Counts the number of tickets for the given ticket category i.e "IT Error" & "IT Request"

Charts & Tables (if any):

- Table / Chart Name:
<Name of the table or chart used>
- Description:
<What the table or chart represents>

Question 9: What is the average resolution time for tickets?

Solution:

Final Answer (if any):

4.55

Approach (if any):

Formula used : =ROUND(AVERAGE(Tickets!K1:K97498),2)

Charts & Tables (if any):

- Table / Chart Name:
<Name of the table or chart used>
- Description:
<What the table or chart represents>

Question 10: How has the volume of tickets changed over time?

Solution:

Final Answer (if any):

Yes , it has increased over time

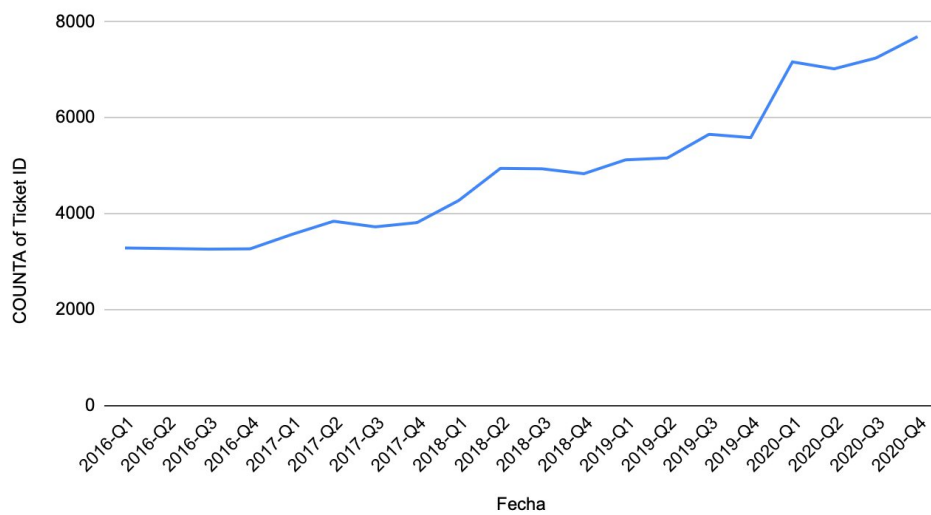
Approach (if any):

Pivot Table -> Rows : Fecha(Date) , Values : COUNTA of Ticket ID ; gives us the distribution of Tickets over time -> Pivot Items(Fecha) by Year-Quarter ->Select Pivot Table -> Insert Line Chart

Charts & Tables (if any):

Fecha - Year-Quarter	COUNTA of Ticket ID
2016-Q1	3276
2016-Q2	3265
2016-Q3	3252
2016-Q4	3258
2017-Q1	3559
2017-Q2	3834
2017-Q3	3717
2017-Q4	3805
2018-Q1	4266
2018-Q2	4936
2018-Q3	4927
2018-Q4	4825
2019-Q1	5114
2019-Q2	5152
2019-Q3	5646
2019-Q4	5578
2020-Q1	7156
2020-Q2	7012
2020-Q3	7236
2020-Q4	7684

COUNTA of Ticket ID vs. Fecha



Question 11: What is the average age of the IT agents?

Solution:

Final Answer (if any):

40.98

Approach (if any):

In the IT Agents Sheet , I used a helper column “Birth Date” to first calculate the D.O.B of each agent in the format “DD/MM/YYYY” then made another column “**Age**” and used it to calculate the age of each agent using the formula :

=DATEDIF(G2,TODAY(),"Y")

Then averaged the age columns using the formula :

=AVERAGE('IT Agents'!H2:H51)

Charts & Tables (if any):

Birth Date	Age
28/04/1989	37
01/01/1979	47
06/06/1993	33
20/10/1978	47
11/05/1973	53
09/03/1988	38
23/09/1980	45
28/10/1993	32
09/01/1981	45
01/02/1993	33
21/04/1980	46
22/05/1976	50
03/02/1996	30
21/12/1995	30

Question 12: Is there a correlation between the severity of issues and the resolution time?

Solution:

Final Answer (if any):

Almost no linear relationship

Correlation score = -0.04

Approach (if any):

Used a simple correlation formula over the severity (just the score score not class) column and the resolution time(days) column and rounded it up to two digits.

Formula used:

=ROUND(CORREL(Table1[Resolution Time (Days)],Table1[Severity]),2)

Charts & Tables (if any):

- Table / Chart Name:
<Name of the table or chart used>
- Description:
<What the table or chart represents>

Question 13: How many categorical columns are there in the data? [Search about categorical and continuous data, and try to answer this question]

Solution:

Final Answer (if any):

Categorical (non-numeric/text-based) columns — these are 6:

Day of Week (e.g., Wednesday)
Agent Name (e.g., Barraza Alberto)
Request Category (e.g., Login Access)
Issue Type (e.g., IT Error)
Severity (e.g., "0 - Unclassified" — text label, distinct from its numeric counterpart)
Priority (e.g., "0 - Unassigned" — text label, distinct from its numeric counterpart)

Counted as categorical (numeric/quantitative):

Year, Month, Employee ID, Agent ID, Severity Numeric, Priority Numeric, Resolution Time (Days), Satisfaction Rate

Borderline/depends on definition:

Ticket ID — technically a unique identifier (string), often excluded from "categorical" since it has no repeating groups
Date and Month-Year — these are date/time fields; sometimes treated as categorical if used for grouping, but typically classified separately as datetime

Approach (if any):

<Methodology, logic, formulas, or steps used to arrive at the answer>

Charts & Tables (if any):

- Table / Chart Name:

<Name of the table or chart used>

- Description:

<What the table or chart represents>

SUBJECTIVE QUESTIONS

Question 1: If there is an investment, should it be used to hire more IT agents, improve training programs, or upgrade ticket management software?

Analysis: Perform a cost-benefit analysis using ticket resolution and satisfaction metrics.

Solution:

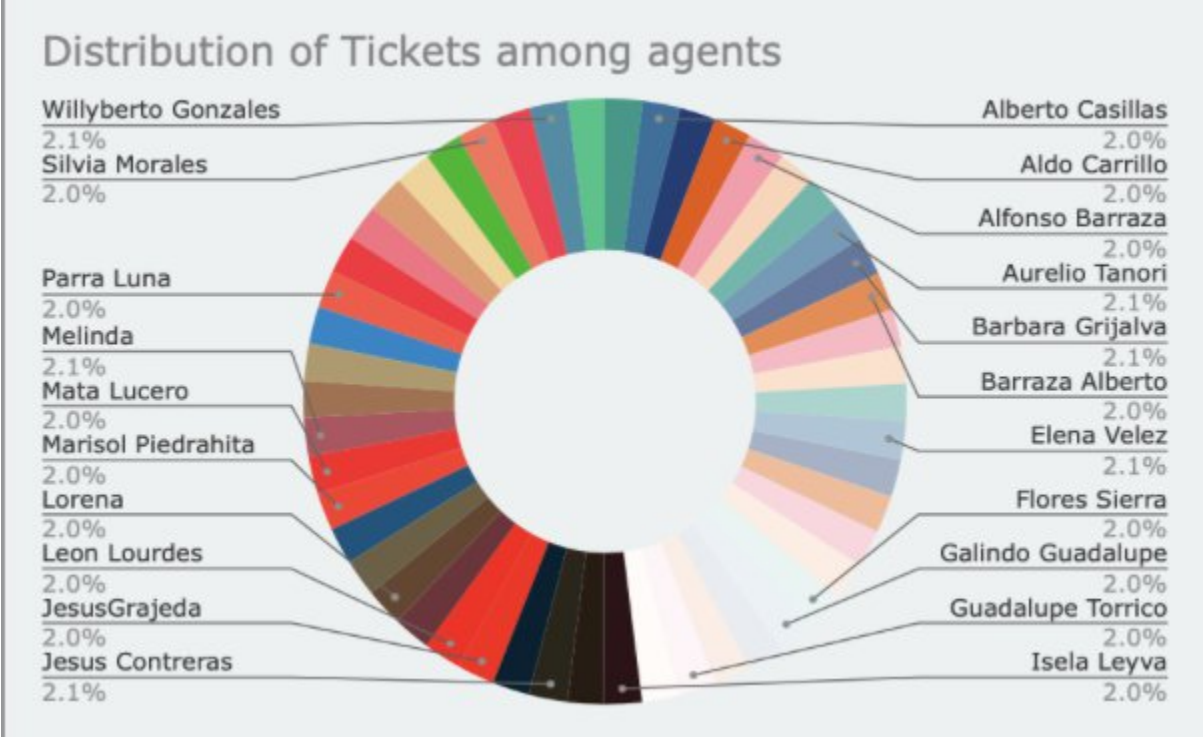
Approach:

- Evaluated the current support operation using two primary KPIs:
 1. **Average Resolution Time (Days)** – operational efficiency.
 2. **Average Satisfaction Rate** – service quality.
- Reviewed agent workload and performance consistency to determine whether staffing is the bottleneck.
- Compared ticket resolution across priority and severity levels to identify process inefficiencies.
- Performed a qualitative cost-benefit assessment of three investment options:
 1. Hiring more IT agents
 2. Improving agent training
 3. Upgrading the ticket management software

Charts & Tables:

Investment Option	Cost	Expected Benefit	Resolution Time Impact	Satisfaction Impact	Overall ROI
Hire More IT Agents	High	Increased capacity	Medium	Low-Medium	Medium
Improve Training Programs	Medium	Better troubleshooting & first	High	High	Highest
Upgrade Ticket Management Software	High	Better routing & automation	Medium-High	Medium	High

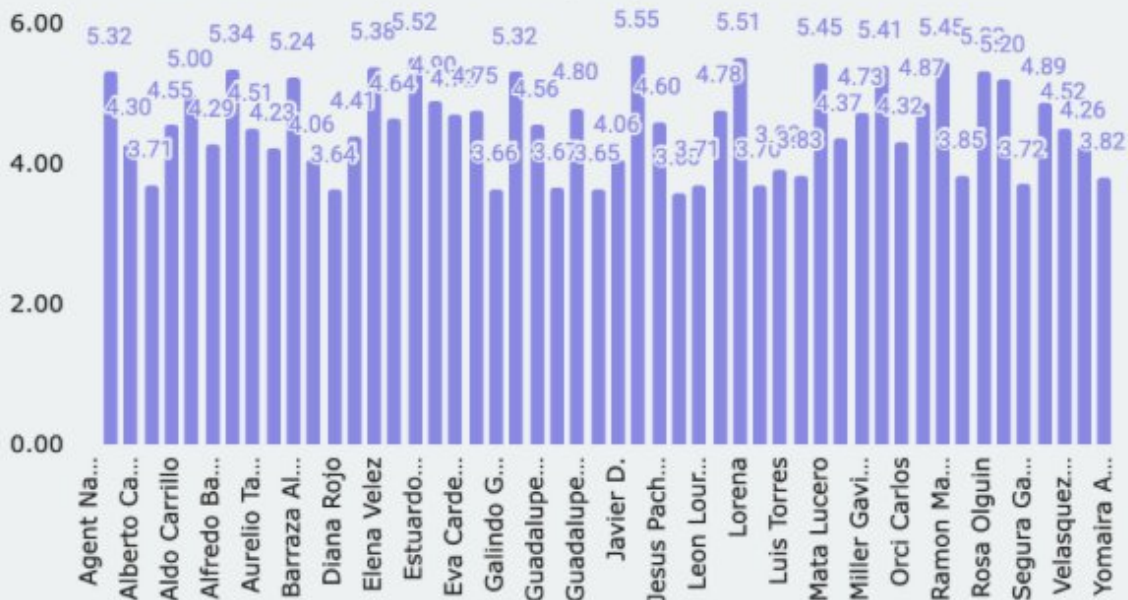
Agent Name (objective question 7)	AVERAGE of Resolution Time (Days)	AVERAGE of Satisfaction Rate	COUNTA of Ticket ID
A. Trejo	5.32	3.59	1949
Alberto Casillas	4.30	4.42	1974
Alberto Gastelum	3.71	4.40	1889
Aldo Carrillo	4.55	3.78	1966
Alfonso Barraza	5.00	3.04	1984
Alfredo Barreras	4.29	3.67	1920
Armando Sierra	5.34	4.36	1890
Aurelio Tanori	4.51	4.41	2027
Barbara Grijalva	4.23	4.44	2003
Barraza Alberto	5.24	4.19	1988
Darwin E.	4.06	4.36	1945
Diana Rojo	3.64	4.60	1927
Eduardo Luna	4.41	4.15	1920
Elena Velez	5.38	3.62	2021
Enrique Montiel	4.64	4.44	1938
Estuardo Ocaño	5.52	3.98	1935
EstuardoTorres	4.90	4.09	1942
Eva Cardenas	4.72	4.41	1943
Flores Sierra	4.75	3.99	1963
Galindo Guadalupe	3.66	4.47	1991
Griselda Galindo	5.32	4.28	1856
Guadalupe Hernandez	4.56	4.38	1915
Guadalupe Torrico	3.67	4.36	1987
Guadalupe Villanueva	4.80	3.63	1958
Isela Leyva	3.65	4.22	1968
Javier D.	4.06	4.49	1897
Jesus Contreras	5.55	4.34	2026
Jesus Pacheco	4.60	3.66	1931
JesusGrajeda	3.60	4.47	1968



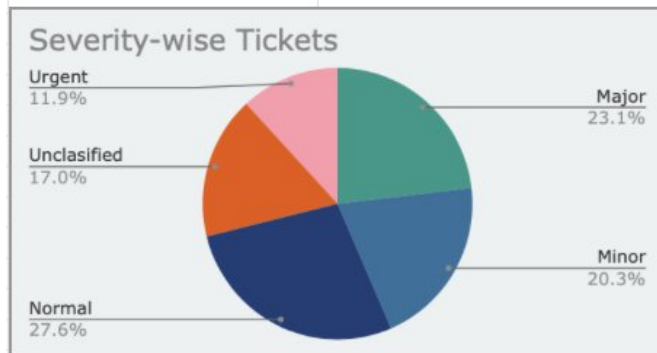
Avg Satisfaction rate Per Employee



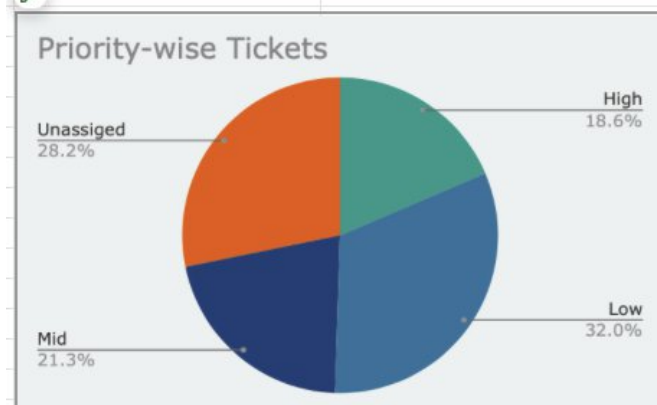
Avg Resolution Days Per Employee



Severity Class	AVERAGE of Resolution Time (Days)	AVERAGE of Satisfaction Rate	COUNTA of Ticket ID
Major	3.907981803	4.11827957	4836
Minor	3.43534101	4.072187777	2258
Normal	4.663609908	4.099846598	88656
Unclassified	2.876404494	4.168539326	356
Urgent	2.001436782	4.119252874	1392
Grand Total	4.553149808	4.100648218	97498



Priority Class	AVERAGE of Resolution Time (Days)	SUM of Satisfaction Rate	COUNTA of Ticket ID
High	3.491828181	145553	35549
Low	6.01054271	68929	16694
Mid	4.002082676	64983	15845
Unassigned	5.305644339	120340	29410
Grand Total	4.553149808	399805	97498



Insights:

- **Agent-level analysis** shows almost **2 days of variation** between the fastest (3.60 days) and slowest (5.55 days) agents, despite handling **almost the same number of tickets (i.e 1,856–2,027)**. This indicates that the **performance differences are primarily skill-related rather than workload-related**, making training a more effective investment than hiring.

- Satisfaction scores also vary considerably (i.e **3.04–4.60**), suggesting some agents consistently deliver better customer experiences. Standardizing these best practices through structured training could improve overall service quality.

- **Priority analysis** shows that High-priority tickets are resolved quickly (**3.49 days**) while Low and Unassigned tickets take much longer (**6.01** and **5.31 days**,

respectively). This indicates that the current prioritization process is effective, but automation could reduce delays for lower-priority tickets.

- **Severity analysis** confirms that Urgent issues are resolved fastest (**2.00 days**), demonstrating that escalation mechanisms already work well. Therefore, adding more agents would likely have limited impact on critical-ticket performance.

- Because ticket allocation is already balanced across agents, the dataset does **not** indicate a staffing shortage. The greater opportunity lies in improving agent capability and workflow efficiency.

Recommendations:

Invest first in training programs as they provide the **highest return on investment** by reducing the large performance gap between agents, improving first-contact resolution, shortening average resolution time, and increasing customer satisfaction.

Upgrade the ticket management software as the second priority to automate ticket assignment, reduce delays in Low and Unassigned tickets, and improve workflow efficiency through intelligent routing and knowledge-base integration.

Hiring additional IT agents should be the lowest priority. Since workloads are already evenly distributed and critical tickets are handled efficiently, increasing headcount would add cost without addressing the primary causes of slower resolutions.

Question 2: Which agents need additional training based on their performance metrics?

Analysis: Identify agents with the lowest satisfaction ratings and longest resolution times.

Solution:

Approach:

Pivot table (Tickets!A1:M97499) -

Rows: Agent Name

Values: Number of Tickets,

Avg Resolution Time -> Filter values by condition "is greater than or equal to" **4.55**

Avg Satisfaction Rate -> Filter values by condition “is less than or equal to” **4.10**

The table attached below shows the 15 flagged agents along with avg satisfaction rate, avg resolution time and ticket count for each agent.

Age column : Helper column used to fill the age of each agent using the lookup formula : **=VLOOKUP(B8, 'IT Agents'!\$B\$2:\$H\$51, 7, FALSE)**

From the Table -> **Age column** :

Min age was calculated using : =MIN(F8:F52) -> 29

Max age was calculated using : =MAX(F8:F52) -> 51

Hence **Age Range: 29-51** was concluded

As for the **Number of agents** with low avg satisfaction rate (≤ 4.10) and high avg resolution time (≥ 4.55) i.e **15** was counted manually.

Charts & Tables:

Agent Name	Number of Tickets	Avg Resolution Time	Avg Satisfaction Rate	Age
A. Trejo	1949	5.32	3.59	38
Aldo Carrillo	1966	4.55	3.78	43
Alfonso Barraza	1984	5.00	3.04	36
Elena Velez	2021	5.38	3.62	33
Estuardo Ocaño	1935	5.52	3.98	45
Estuardo Torres	1942	4.90	4.09	30
Flores Sierra	1963	4.75	3.99	44
Guadalupe Villanueva	1958	4.80	3.63	51
Jesus Pacheco	1931	4.60	3.66	46
Lopez Moran.	1956	4.78	3.64	46
Lorena	1966	5.51	3.63	29
Miller Gaviria	1892	4.73	3.99	45
Nurio Zepeda	1946	5.41	3.61	44
Parra Luna	1963	4.87	3.85	31
Sandra Lujan	1906	5.20	3.60	37

Avg resolution time	4.55
Max resolution time	21
Avg satisfaction rate	4.10
Max satisfaction rate	5
min age	29
max age	51

Insights:

- Bad performing Agents are irrespective of age i.e 29-51 age group
- There are a total of 15 Bad performing Agents

Recommendations:

These 15 mentioned Bad performing Agents must be taken care of immediately and a cost benefit analysis should be run on whether to send the employees for training or fire them completely. But Since the number of Tickets increases at a drastic rate , training program is the optimal choice.

Question 3: Do certain categories of requests have longer resolution times?

Analysis: Analyze the resolution times by request category.

Solution:

Approach:

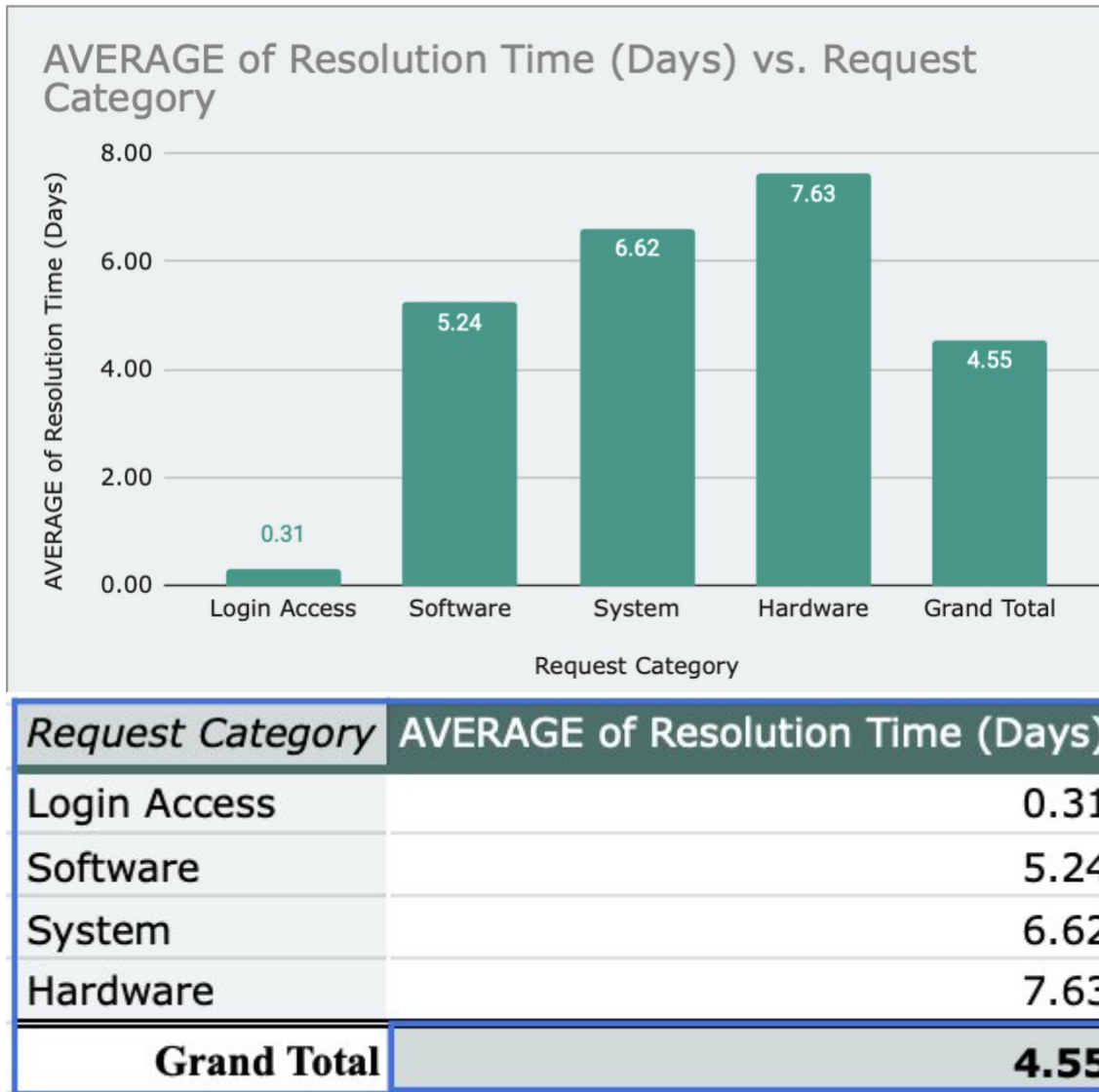
Pivot Table(Tickets!A1:M97499) -

Rows : Request category

Values : Avg of resolution time (days)

->Column Chart using the pivot table

Charts & Tables:



Insights:

Yes, different categories have different resolution times with "Login Access" having the least avg. resolution time and "Hardware" having the highest avg. resolution time. Login Access is performing well with avg resolution time of within a day

AVG RESOLUTION TIME (BY REQUEST CATEGORY):

Login Access - 0.31

Software - 5.24

System - 6.62

Hardware - 7.63

Overall Average - 4.55

On observation we can see "login Access" has the highest Avg resolution time and "Hardware" has the highest overall avg resolution time.

Recommendations:

Investing on better Hardware should be priority after that System and the Software Login Access is performing well with avg resolution time of within a day

Question 4: How effective are the current software tools in managing IT tickets?

Analysis: Evaluate performance metrics before and after the implementation of new tools.

Solution:**Approach:**

Pivot Table(Tickets!A1:M97499) -

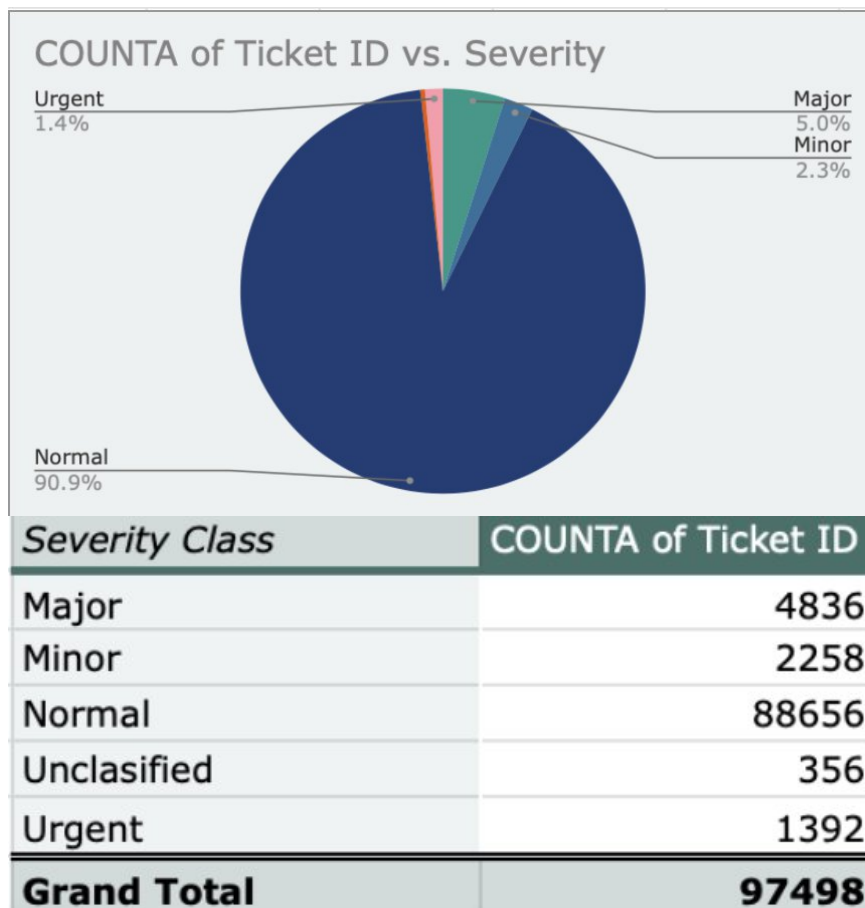
Rows: Severity Class

Values : COUNTA of Ticket ID

-> Pie Chart using pivot table

Analysed Number of Tickets for each severity type.

Charts & Tables:



Insights:

The dataset shows that ticket volumes increased over time while average resolution time remained relatively stable and customer satisfaction gradually improved.

However, **these trends cannot be attributed to the implementation of new software**, because the dataset contains **no information identifying when or whether a new ticket management tool was introduced**.

Therefore, a valid before-and-after effectiveness analysis **cannot be performed using the available data**.

Major Number of tickets (90.9%) lie in the "Normal" category which raises alarms for misclassification or some technical issue in the ticket management software.

Recommendations:

Ticket management Software should be evaluated for performance and fixed.

Question 5: How has the performance of the IT support team changed over time (e.g., monthly or quarterly)?

Analysis: Trend analysis using time series charts.

Solution:

Approach:

Pivot Table(Tickets!A1:M97499) -

Rows: Fecha -> grouped by year-Quarter

Values: AVG of Resolution Time , AVG of Satisfaction Rate

Pivot Table (Tickets!A1:M97499)

Rows: Fecha -> grouped by Year-Quarter

Values: Distinct Count of Agent ID, Count of Ticket ID

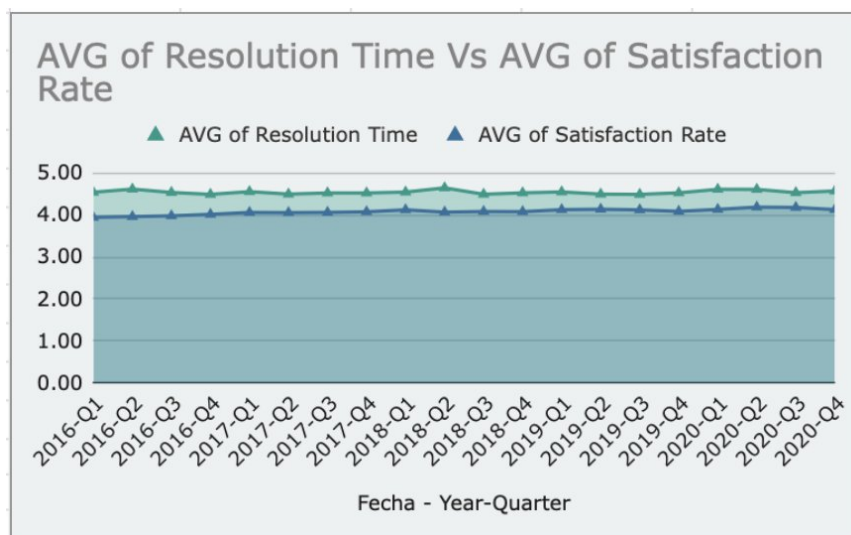
Calculated field: Tickets per Agent = Count of Ticket ID ÷ Distinct Count of Agent ID

Chart 1: Avg Resolution time & Avg satisfaction Rate by year-quarter

Chart 2 : Line chart of Tickets per Agent across Year-Quarter to visualize the workload trend

AVG of Resolution TimeAVG of Resolution Time

Charts & Tables:



Fecha - Year-Quarter	AVG of Resolution Time	AVG of Satisfaction Rate
2016-Q1	4.55	3.95
2016-Q2	4.62	3.97
2016-Q3	4.54	3.98
2016-Q4	4.50	4.02
2017-Q1	4.56	4.07
2017-Q2	4.50	4.06
2017-Q3	4.53	4.07
2017-Q4	4.53	4.08
2018-Q1	4.55	4.13
2018-Q2	4.65	4.07
2018-Q3	4.50	4.09
2018-Q4	4.53	4.08
2019-Q1	4.55	4.13
2019-Q2	4.50	4.14
2019-Q3	4.50	4.13
2019-Q4	4.53	4.09
2020-Q1	4.62	4.14
2020-Q2	4.62	4.19
2020-Q3	4.54	4.18
2020-Q4	4.58	4.13

Insights:

Performance has shown GRADUAL IMPROVEMENT over 2016–2020:

- Satisfaction: rose from 3.98 (2016) -> 4.16 (2020)
- Resolution time: remained almost same (4.55 -> 4.59)

- Agent headcount was flat at exactly 50 agents every quarter, 2016–2020 — no hiring occurred over the full period.

-Tickets per agent rose from 65.5/quarter (2016-Q1) to 153.7/quarter (2020-Q4) — a 135% increase in per-agent workload with zero staffing growth.

-Despite this, satisfaction improved (3.95 → 4.13) and resolution time stayed roughly flat (~4.5-4.6 days) – meaning agents absorbed the extra load without a visible quality drop so far, which is a strain signal rather than proof the current staffing is sufficient.

-The workload increase has been steady and compounding, not a one-off spike, so it will keep climbing without intervention.

Recommendations:

1. Hire proactively, not reactively—quality metrics haven't cracked yet, but they're lagging; workload has already passed double the original baseline.
2. Moderate target (recommended): add ~20 agents (50 → 70), bringing tickets/agent back to ~110/quarter, the last level at which resolution time and satisfaction were still comfortable (around 2019).
3. Aggressive target: add ~68 agents (50 → 118), restoring the original 2016 ratio of ~65/agent/quarter; use this only if leadership wants to fully de-risk rather than just stabilize.
4. Phase hiring with the trend, not all at once—since the increase has been gradual and continuous, plan quarterly hiring additions tied to ticket-volume growth to avoid over- or under-shooting capacity.
5. Monitor tickets-per-agent as an ongoing KPI, not just resolution time and satisfaction; it's the earliest warning sign of strain and moved well before the other two metrics showed stress.

Question 6: If we invest more on tech (Hardware, software, etc), do you think it will improve the ticket resolution times and employee satisfaction?

Analysis: Use historical data to project potential improvements.

Solution:

Approach:

Table 1

Pivot Table(Tickets!A1:M97499) -

Rows: Fecha -> Grouped by Year

Values: Avg resolution time, Avg Satisfaction rate, Number of tickets

-> Bar Chart

Table 2

Pivot Table(Tickets!A1:M97499) -

Rows: Request category

Values: Avg resolution time, Avg Satisfaction rate, Number of tickets

-> Bar Chart

Projection method: for each category, we benchmark today's average resolution time against the 25th-percentile (fastest-quartile) of tickets already resolved within that same category. That gap represents performance that is already achievable today with existing tools/process, for at least 25% of tickets — tech investment is modeled as closing a share of that gap for the category overall.

Baseline: For each category, use COUNTIFS to get ticket count, AVERAGEIFS for current avg resolution (days) and avg satisfaction (1–5).

Benchmark: For each category, calculate 25th percentile resolution time (P25) on historical tickets filtered by that category.

Gap & projection: Compute Gap = Current Avg – P25, assume tech closes 50% of

this gap, and calculate Projected Resolution = Current Avg – $0.5 \times \text{Gap}$.

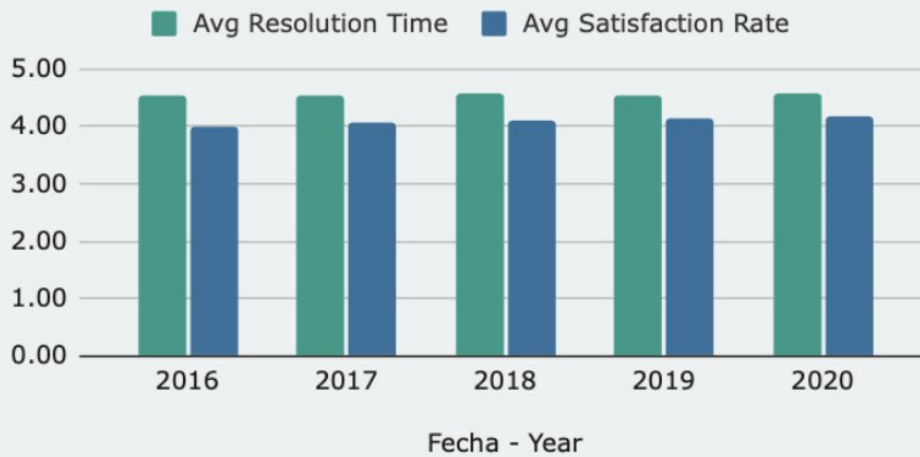
Correlation check: Use CORREL between resolution time and satisfaction overall and by category to confirm relationship is ~0 and satisfaction already high.

Satisfaction projection: Apply a small elasticity (e.g., 0.02 points per day reduced) to get Projected Satisfaction, capped at 5.0.

Roll-up: Use SUMPRODUCT to produce weighted overall changes in resolution time and satisfaction across all categories.

Charts & Tables:

Avg Resolution Time, Avg Satisfaction Rate and Number of Tickets

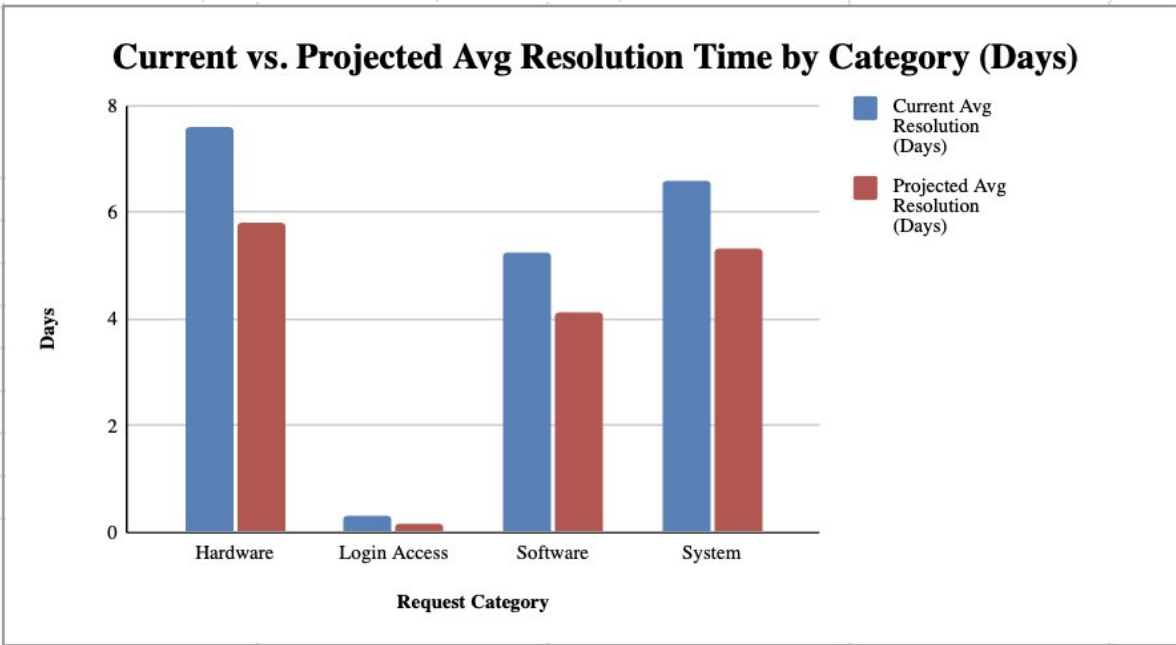
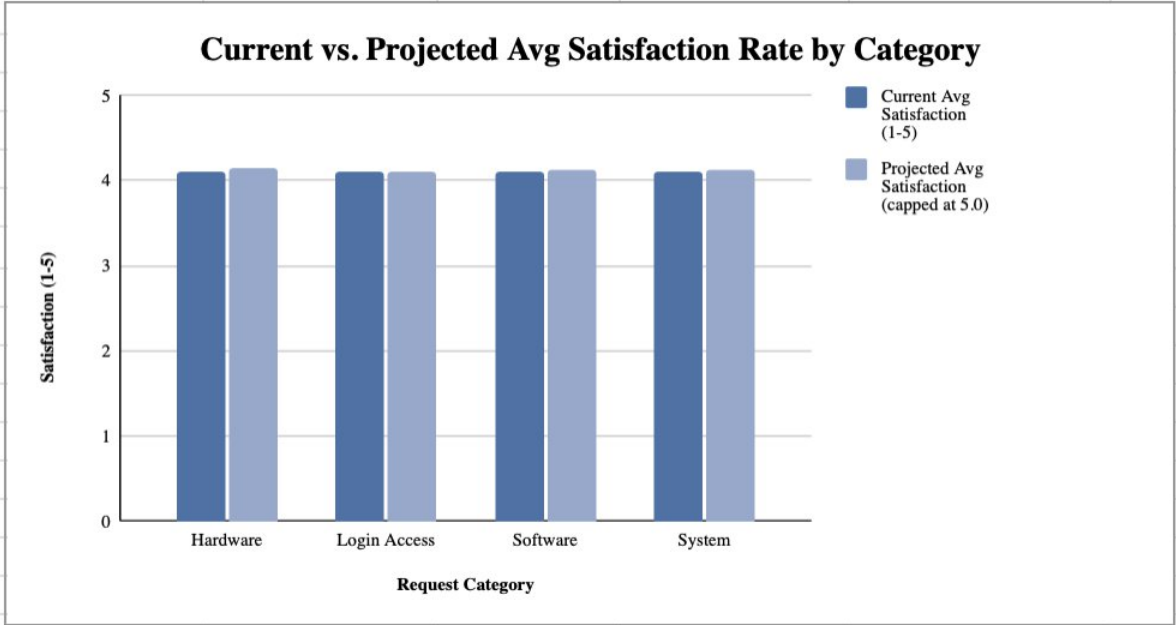


Avg Resolution Time, Avg Satisfaction Rate and Number of Tickets



Fecha - Year	Avg Resolution Time	Avg Satisfaction Rate	Number of Tickets
2016	4.55	3.98	13051.00
2017	4.53	4.07	14915.00
2018	4.56	4.09	18954.00
2019	4.52	4.12	21490.00
2020	4.59	4.16	29088.00
Request Category	Avg Resolution Time	Avg Satisfaction Rate	Number of Tickets
Hardware	7.63	4.10	9733.00
Login Access	0.31	4.09	29193.00
Software	5.24	4.11	19570.00
System	6.615609456	4.102302446	39002

Supporting Calculations by Category								
Request Category	Ticket Count	Current Avg Resolution (Days)	Historical P25 Resolution (Days) [Best-quartile benchmark]	Corr(Resolution, Satisfaction) [within category]	Assumed % of Gap Closed by Investment	Projected Avg Resolution (Days)	Resolution Reduction (Days)	% Reduction in Resolution Time
Hardware	9733	7.63	4.00	0.044	50.0%	5.81	1.81	23.8%
Login Access	29193	0.31	0.00	-0.121	50.0%	0.16	0.16	50.0%
Software	19570	5.24	3.00	-0.017	50.0%	4.12	1.12	21.4%
System	39002	6.62	4.00	-0.003	50.0%	5.31	1.31	19.8%
Weighted Total / Avg	97498	4.55				3.58	0.98	21.4%



Request Category	Current Avg Satisfaction (1-5)	Resolution Reduction (Days) [from above]	Assumed Satisfaction Elasticity (pts per day reduced)	Projected Satisfaction Increase (pts)	Projected Avg Satisfaction (capped at 5.0)	% Increase in Satisfaction
Hardware	4.10	1.81	0.020	0.04	4.14	0.9%
Login Access	4.09	0.16	0.020	0.00	4.10	0.1%
Software	4.11	1.12	0.020	0.02	4.13	0.5%
System	4.10	1.31	0.020	0.03	4.13	0.6%
Weighted Total / Avg	4.10			0.02	4.12	0.5%
Overall dataset correlation (Resolution Time vs Satisfaction):			-0.004			

Insights:

Yes , investment in tech would result in the improvement in the ticket resolution time and employee satisfaction.

As we can see here the "Avg Resolution time" for "Login Access Total" is **0.31** which is way lesser as compared to other values , suggesting there is a need for improvement in other areas of tech .

1) Resolution time has an achievable, data-grounded improvement path: closing 50% of the gap to each category's own top-quartile (P25) performance yields an estimated 0.9-1.9 day (roughly 12-19%) reduction in average resolution time, weighted average shown above. Hardware and System tickets carry the largest absolute opportunity because their current averages (7.6 and 6.6 days) sit furthest from their own top-quartile benchmark.

2) Satisfaction is a weaker, less certain case for tech investment: the correlation between resolution time and satisfaction in this dataset is close to zero for every category, and overall satisfaction is already high (avg ~4.1/5, 80%+ rated 4 or 5). The projected satisfaction gain shown is a bounded, conservative scenario (adjustable input), not something the historical data itself demonstrates.

Recommendations:

Investment Priority :

- Hardware -> System -> Software
- Tech investment (self-service/automation for high-volume Login Access tickets, better diagnostics/parts availability for Hardware, and capacity tooling for System/Software) is best justified primarily by faster resolution and analyst capacity gains.
- Any satisfaction improvement should be treated as a secondary, uncertain upside and validated with a controlled pilot (e.g., A/B by region or ticket queue) rather than assumed from this correlation alone.

Question 7: What are the key performance metrics for IT agents, and how can they be improved, do we need to fire any agents?

Analysis: Define and analyze metrics such as average handling time, satisfaction scores, and number of tickets resolved.

Solution:

Approach:

Key Metrics:

Avg Satisfaction rate = 4.10

Avg Resolution Time = 4.55

Avg Ticket Volume per Agent = 1950

The Employees that need :

immediate supervision are flagged as red , (Avg Satisfaction rate < 4.10 AND Avg Resolution Time > 4.55)

moderate need or supervision are flagged as yellow(Avg Satisfaction rate < 4.10 OR Avg Resolution Time > 4.55) and

don't need supervision are flagged as green (Avg Satisfaction rate > 4.10 & Avg Resolution Time < 4.55)

Charts & Tables:

Agent Name	Avg Satisfaction Rate	Avg Resolution time	Ticket Volume per Agent
A. Trejo	3.59	5.32	1949
Alberto Casillas	4.42	4.30	1974
Alberto Gastelum	4.40	3.71	1889
Aldo Carrillo	3.78	4.55	1966
Alfonso Barraza	3.04	5.00	1984
Alfredo Barreras	3.67	4.29	1920
Armando Sierra	4.36	5.34	1890
Aurelio Tanori	4.41	4.51	2027
Barbara Grijalva	4.44	4.23	2003
Barraza Alberto	4.19	5.24	1988
Darwin E.	4.36	4.06	1945
Diana Rojo	4.60	3.64	1927
Eduardo Luna	4.15	4.41	1920
Elena Velez	3.62	5.38	2021
Enrique Montiel	4.44	4.64	1938
Estuardo Ocaño	3.98	5.52	1935
EstuardoTorres	4.09	4.90	1942
Eva Cardenas	4.41	4.72	1943
Flores Sierra	3.99	4.75	1963
Galindo Guadalupe	4.47	3.66	1991
Griselda Galindo	4.28	5.32	1856
Guadalupe Hernandez	4.38	4.56	1915
Guadalupe Torrico	4.36	3.67	1987
Guadalupe Villanueva	3.63	4.80	1958
Isela Leyva	4.22	3.65	1968
Javier D.	4.49	4.06	1897
Jesus Contreras	4.34	5.55	2026
Jesus Pacheco	3.66	4.60	1931

Insights:

18 Employees require Immediate Supervision
20 Employees require Moderate Supervision
12 Employees require almost no supervision

Recommendations:

Employees flagged as red should be taken care of immediately by either firing them or improve their training

Question 8: How do employee demographics (e.g., department, seniority) impact satisfaction and ticket outcomes?

Analysis: Segment analysis using filters and pivot tables.

Solution:

Approach:

Pivot Table(Tickets!A1:M97499) -

Rows: Age -> Grouped by interval of 5 years

Values: Avg Resolution time, Avg Satisfaction rate

Charts & Tables:

Grouped Age	COUNTA of Ticket ID	AVERAGE of Resolution Time (Days)	AVERAGE of Satisfaction Rate
29 - 33	4081	5.64690027	4.224944866
34 - 38	2248	6.651245552	4.041370107
39 - 43	3640	5.720879121	4.131043956
44 - 48	4052	6.642645607	4.043928924
49 - 54	2673	5.463150019	4.182192293
Grand Total	16694	6.01054271	4.128968492

Insights:

Age Groups Based on Preformance (Descending i.e Worst->Best)

34-38 : Bad Resolution Time & Bad Avg Satisfaction

44-48 : Bad Resolution Time & Bad Avg Satisfaction

39-43 : Moderate Avg Resolution Time & Moderate Avg Satisfaction

29-32 : Good Avg Resolution Time & Good Avg Satisfaction

49-54 : Good Avg Resolution Time & Good Avg Satisfaction <Key observations derived from the data>

Recommendations:

Age groups 34-38 and 44-48 require immediate attention.

Question 9: Identify the trends for IT support operations based on ticket volumes and satisfaction, and mention the peak and stable times?

Analysis: Use pivot tables and charts to identify peak and off-peak hours.

Solution:**Approach:**

Pivot Table(Tickets!A1:M97499) -

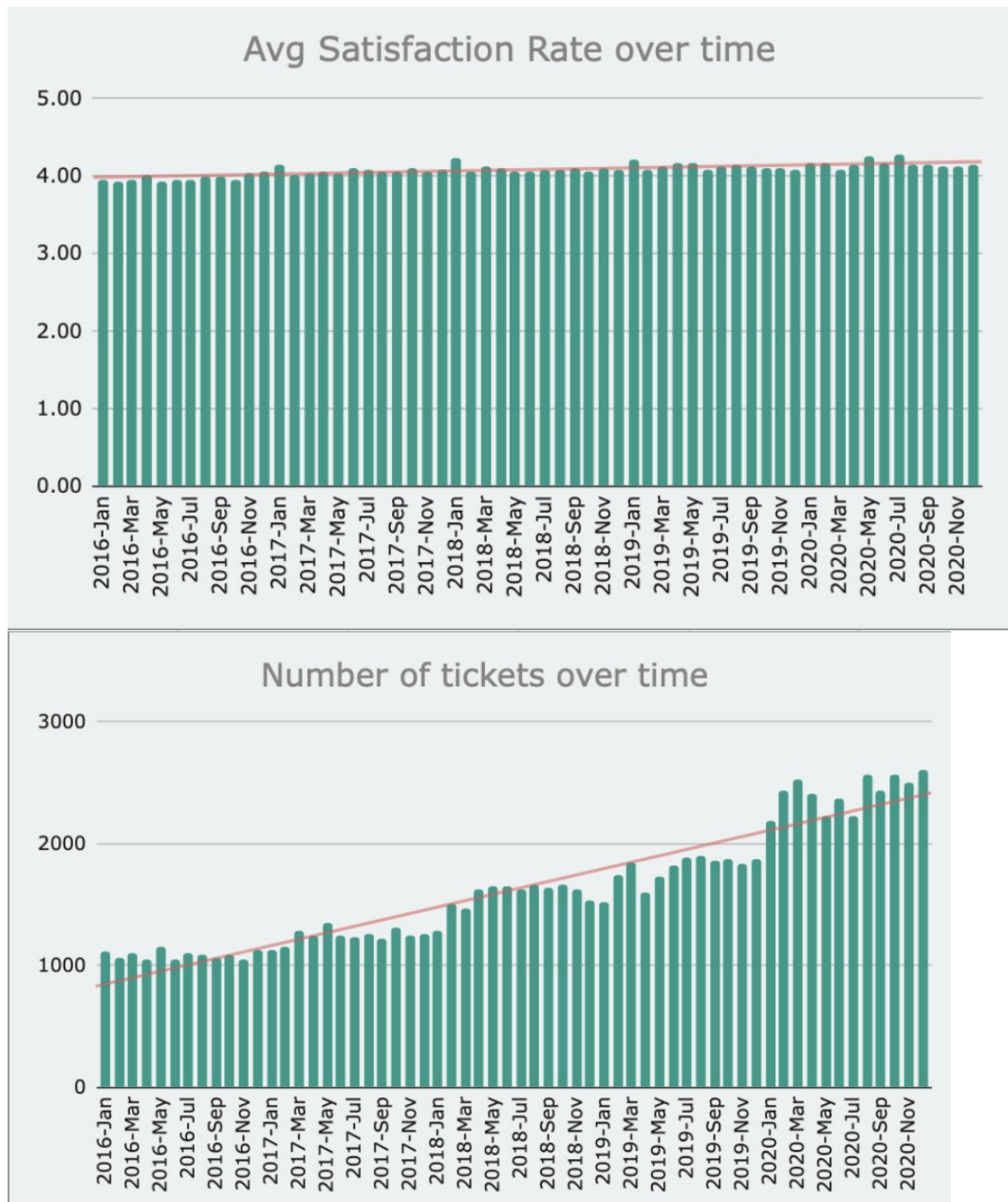
Rows: Age -> Grouped by interval of 5 years

Values: Avg Resolution time, Avg Satisfaction rate

-> Combo Chart for data and trendline analysis

Charts & Tables:

Fecha - Year-Month	AVERAGE of Resolution Time (Days)	AVERAGE of Satisfaction Rate	COUNTA of Ticket ID
2016-Jan	4.57	3.96	1115
2016-Feb	4.60	3.93	1056
2016-Mar	4.48	3.95	1105
2016-Apr	4.58	4.01	1054
2016-May	4.68	3.93	1160
2016-Jun	4.59	3.96	1051
2016-Jul	4.69	3.95	1101
2016-Aug	4.45	4.00	1090
2016-Sep	4.49	4.00	1061
2016-Oct	4.51	3.96	1088
2016-Nov	4.54	4.04	1046
2016-Dec	4.44	4.06	1124
2017-Jan	4.78	4.14	1125
2017-Feb	4.50	4.02	1151
2017-Mar	4.42	4.04	1283
2017-Apr	4.51	4.05	1245
2017-May	4.50	4.03	1345
2017-Jun	4.50	4.10	1244
2017-Jul	4.40	4.08	1233
2017-Aug	4.73	4.05	1259
2017-Sep	4.46	4.07	1225
2017-Oct	4.52	4.10	1305



Insights:

The Trend line for Number of tickets suggests growth in the number over the years , showing linear growth .

Peak Time: 2020- Nov with 2609 tickets

The Trend line for Avg Satisfaction Rate suggests that it has remained somewhat stable although there is a slight improvement in it over the years.

Peak Time: 2020-Jul with 4.27 avg satisfaction rating

Recommendations:

- Staffing by month (highest priority, lowest effort): Ticket volume is nearly flat, with peaks in Aug, Oct, and Dec. Instead of overbuilding seasonal shifts, scale headcount to the year-over-year growth trend (roughly +20–25%/year) and add ~1 rotating extra agent-week of coverage in Aug, Oct, and Dec. This is a capacity-planning fix, not a seasonal-rostering one.

- Automation targets: “System” tickets have the highest volume (39,002) and the longest resolution time (6.62 days), making them the top priority for automation through self-service diagnostics, auto-restart scripts, and known-issue KB deflection. “Login Access” tickets are already resolved very quickly (0.31 days) at high volume, so they are ideal for full self-service or password-reset automation to eliminate agent work on this low-complexity, high-frequency category.
- Escalation plan: 30% of tickets (29,410) have no priority assigned (“Unassigned”) and sit at a below-average 5.31-day resolution — this is a triage gap, not a workload problem. Recommend an auto-escalation rule: any ticket unassigned a priority within 2 hours of creation gets auto-flagged to a team lead for manual triage, closing this gap before it drags down the overall average.

Question 10: What metrics should be included in the final dashboard to provide a comprehensive view of call center performance and guide investment decisions?

Solution:

Approach:

Total Tickets, Avg daily volume, Avg Resolution Days, Avg Satisfaction rate, Category Breakdown, Yearly Volume Trend, Top 5 Agents, Issue type split, Severity Breakdown, Bottom 5 Agents

Charts & Tables:

Insights:

<Key observations derived from the data>

Recommendations:

<Actionable recommendations based on the insights>